

Polar and Rectangular Forms of Equations

Date_____ Period____

Convert each equation from polar to rectangular form.

1) $\tan \theta = 2$

2) $r = 4\cos \theta - 4\sin \theta$

3) $r = -2\cos \theta$

4) $r = 2\cos \theta + 2\sin \theta$

Convert each equation from rectangular to polar form.

5) $(x - 1)^2 + (y + 1)^2 = 2$

6) $x = y^2$

7) $x = y^2$

8) $y = \frac{x^2}{5}$

Convert each equation from polar to rectangular form.

9) $r = 4\csc \left(\theta + \frac{\pi}{6} \right)$

10) $r = 2\sin \left(\theta + \frac{\pi}{4} \right)$

11) $r^2 = 5\sec (2\theta)$

12) $r^2 = 4\sec (2\theta)$